

INGENIOUS CORE, DEXTEROUS HANDS: A DUAL-ARM INTELLIGENT ROBOTIC SYSTEM INTEGRATING LARGE MODELS AND MULTIMODAL PERCEPTION

ABSTRACT

This project designs and implements an intelligent dual-arm robot system integrating large language models and multimodal perception. Based on the Intel DK-2500 development kit, the system integrates an RGB-D camera, LiDAR, microphone array, dual robotic arms, and a mobile chassis, forming a closed-loop architecture of “ perception - decision-making - execution - feedback. ” YOLOv8 is used for object detection, while depth information is combined to achieve 3D localization. Speech recognition, OCR, sound source localization, and emotion recognition are introduced to enhance multimodal interaction. Large language models, LangChain, and RAG are employed for natural language understanding, knowledge retrieval, and task planning. Test results show that the system can perform voice interaction, object recognition, robotic arm grasping, mobile obstacle avoidance, and knowledge-based question answering, demonstrating good practical application value.

Key words : intelligent robot, large language model, multimodal perception, dual robotic arms, human-robot interaction

1. Introduction

With the rapid development of artificial intelligence, robot control, machine vision, edge computing, and large language models, robots are evolving from mechanical execution devices into intelligent embodied systems. Traditional robots can complete fixed tasks in structured industrial environments, but they still face difficulties in service scenarios where objects, users, lighting conditions, and task requirements change frequently. Therefore, a service robot needs stable multimodal perception, natural interaction, reliable reasoning, and safe physical execution.

The goal of this project is to build a practical intelligent robot that can understand user instructions, perceive surrounding objects, make task decisions, and execute actions through dual arms and a mobile platform. The design emphasizes a complete closed loop from perception to decision-making, from decision-making to execution, and from execution to user feedback.

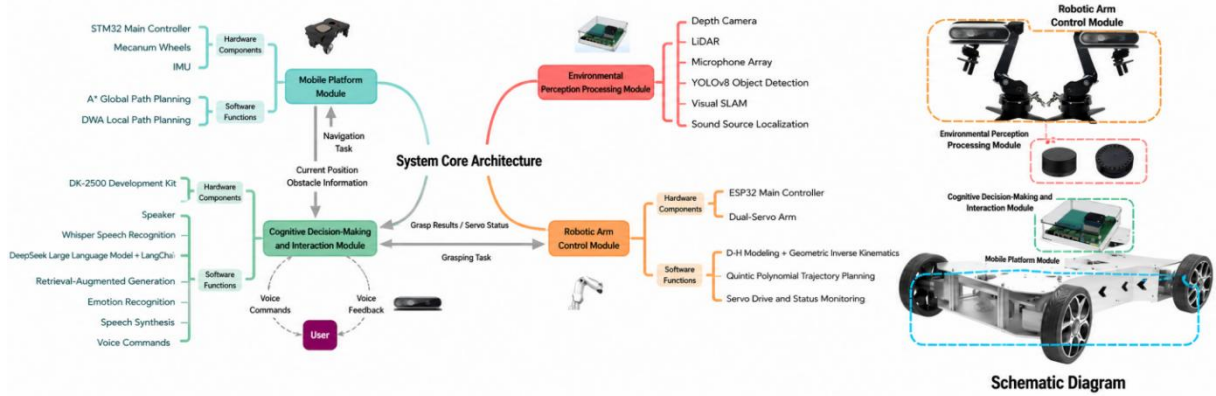


Fig. 1 Module Design Schematic Diagram

2. Overall System Architecture

The system is designed around the idea of “multimodal perception - large-model cognition - precise execution.” The Intel DK-2500 development kit is used as the main computing platform. It is responsible for image processing, depth calculation, speech and text processing, vector retrieval, large-model reasoning, and high-level task scheduling. The lower-level controllers are responsible for robotic arm motion, chassis movement, and sensor data acquisition. The modules exchange commands and status information through a layered embedded architecture.

The robot includes four major modules. The environmental perception module collects RGB images, depth maps, LiDAR data, voice signals, sound direction, and text information. The cognitive interaction module interprets user instructions, retrieves local knowledge, plans task steps, and generates structured commands. The robotic arm control module performs coordinate transformation, inverse kinematics, and trajectory planning. The mobile platform module realizes chassis control, indoor navigation, and real-time obstacle avoidance.

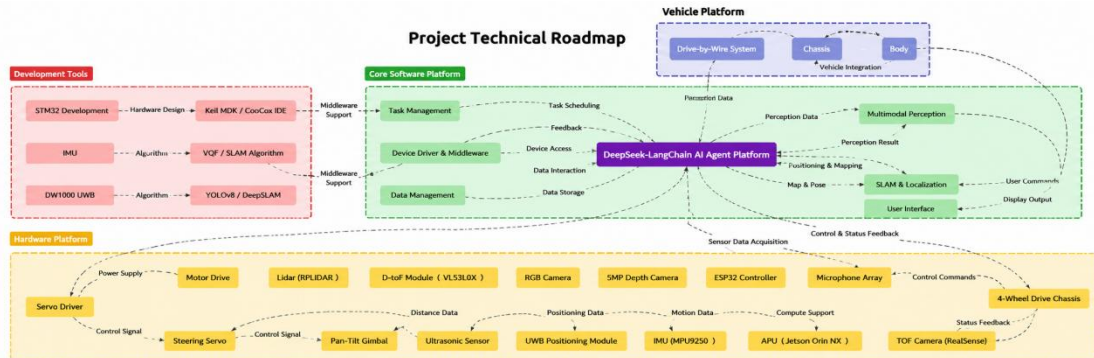


Fig. 2 Project Technical Roadmap

2.1 Hardware Platform and Feasibility

- Intel DK-2500 development kit: provides edge computing capability for AI inference, sensor fusion, and task scheduling.
- RGB-D camera: obtains color images and depth maps for object recognition and 3D target localization.
- LiDAR and microphone array: support obstacle perception, navigation, and sound source localization.
- Dual robotic arms and mobile chassis: execute grasping, placing, moving, and service delivery tasks.

The selected hardware platform provides sufficient interfaces for USB cameras, LiDAR, serial communication, and motion controllers. It also supports local deployment of lightweight perception and retrieval modules, reducing network dependence and improving response stability. The modular design allows each subsystem to be tested independently and then integrated into the final closed-loop system.

2.2 System Workflow

When a user gives a command, the speech module converts it into text and sends it to the cognitive module. At the same time, the perception module updates object positions, depth information, obstacle data, and interaction states. The large-model decision module combines the instruction, current robot state, and retrieved knowledge to output a structured action command. After rule validation and safety checking, the command is executed by the arm or chassis controller, and the result is reported to the user through speech feedback.

3. Multimodal Perception Principle and Implementation

Multimodal perception is the basis of autonomous operation. In this system, visual perception, depth perception, acoustic perception, OCR recognition, speech interaction, and emotion recognition are integrated to form a unified environmental understanding module. The vision module adopts YOLOv8 because it can directly predict object categories and bounding boxes in a single forward process, which is suitable for real-time robotic applications. The detection targets include blocks, cylinders, cups, and medicine packages.

After YOLOv8 detects the target in the RGB image, the system obtains the corresponding depth value from the RGB-D camera. According to the camera intrinsic parameters, the pixel coordinate is converted into a 3D point in the camera coordinate system. Then, through external calibration between the camera and the robotic arm base, the point is transformed into the arm coordinate system. To improve reliability, the system uses multi-frame confirmation, confidence filtering, invalid-depth removal, workspace limitation, and inverse-kinematics feasibility checking before executing grasping.

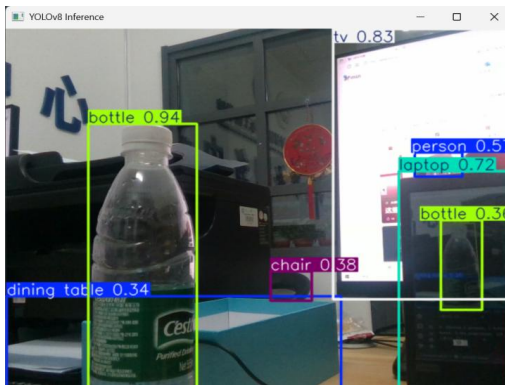


Fig. 3 YOLOv8 object detection result.

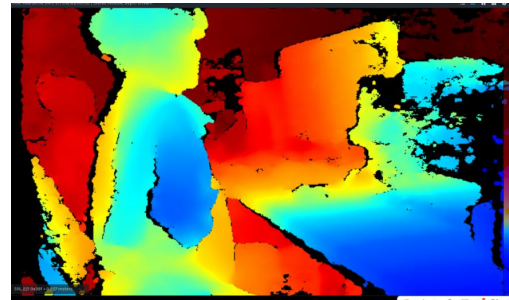


Fig. 4 Depth information used for 3D localization.

3.1 Acoustic, Textual, and Emotional Perception

The microphone array estimates the direction of the speaker using the time difference of arrival model. The generalized cross-correlation with phase transform method is used to estimate time delays between microphone channels, which improves robustness in noisy indoor environments. OCR is used to recognize medicine names, labels, and package information, converting visual text into structured semantic information. Speech recognition and speech synthesis form a voice interaction loop. In addition, MediaPipe-based emotion recognition detects user facial expressions and provides emotional context for response generation.

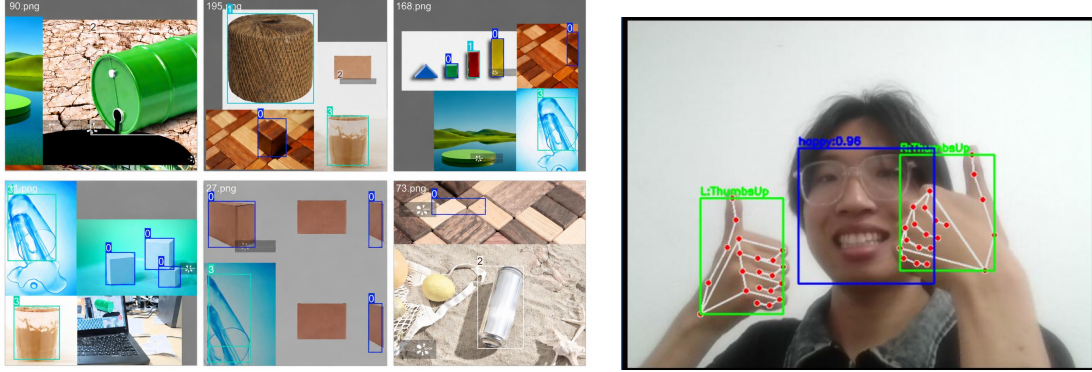


Fig. 5 Object Detection and Face Recognition

These perception results are not isolated outputs. Object categories, target coordinates, recognized text, user voice commands, sound direction, and emotional states are all converted into structured information. The cognitive module can then use them as context for task planning, while the motion module can use target coordinates and workspace constraints for safe execution.

4. Large-Model Cognitive Decision-Making

The cognitive decision-making module is the high-level intelligence center of the robot. Instead of directly connecting large-model output to motors, the system adopts a safe layered strategy: semantic understanding, knowledge retrieval, structured output, rule validation, and action execution. This method keeps the flexibility of natural language understanding while ensuring the controllability of physical actions.

LangChain is used to organize prompt templates, retrieval chains, model calls, and output parsing. The RAG mechanism retrieves relevant information from the local knowledge base before generation. The knowledge base includes allowed actions, task rules, state-machine constraints, medicine information, object descriptions, and safety requirements. FAISS is used to build the vector database for fast semantic search. The retrieved context is combined with the user command and current robot state, and then sent to the large language model for decision-making.

4.1 Structured Command Generation

For robotic execution, the model is required to output a structured command rather than free-form text. The command contains an action name and the reason for selecting the action. The action must belong to the predefined safe action set, such as initialization, object search, right-arm grasping, left-arm receiving, dual-arm pouring, status reporting, task ending, or task abortion. If the output format is invalid, the command is not in the allowed action set, or the current state does not satisfy the precondition, the scheduler rejects the command and returns a safe response.

This design bridges the gap between open-ended language intelligence and deterministic embedded control. It allows users to express instructions naturally, such as “help me pour water” or “find the cup,” while the robot still executes only validated actions. For medicine recognition, the system combines

OCR results, structured CSV matching, and RAG retrieval to generate explanatory answers without providing unsafe dosage or prescription suggestions.

4.2 Humanized Interaction Strategy

Emotion recognition results are used as additional context for interaction. When the user is calm or positive, the robot gives natural confirmation. When sadness, fear, anger, or urgent expressions are detected, the response becomes softer, clearer, and more task-focused. This improves the usability of the robot in care-oriented scenarios.

5. Motion Control and Mobile Platform

The robotic arm control module converts target coordinates into executable joint motions. Each arm is modeled using the Denavit-Hartenberg method. Forward kinematics calculates the end-effector pose from joint angles, while inverse kinematics calculates suitable joint angles from the desired target position. After the target posture is obtained, polynomial trajectory planning is used to generate smooth joint trajectories and reduce sudden changes in velocity.

For dual-arm operation, the system adopts a coordinated strategy. One arm can hold or receive an object while the other arm grasps, transfers, or pours. Before execution, the system checks workspace boundaries, coordinate validity, inverse-solution feasibility, joint feedback, and potential collision constraints. This improves safety during cooperative manipulation and prevents invalid target positions from being sent to the hardware.

5.1 Mobile Navigation and Obstacle Avoidance

The mobile platform enables the robot to move in indoor environments. Based on the chassis kinematic model, high-level motion instructions are converted into wheel-speed commands. LiDAR data are used to detect obstacles around the robot. With global path planning and local obstacle avoidance, the robot can approach the user or target area, avoid obstacles, and cooperate with the robotic arms during object delivery tasks.

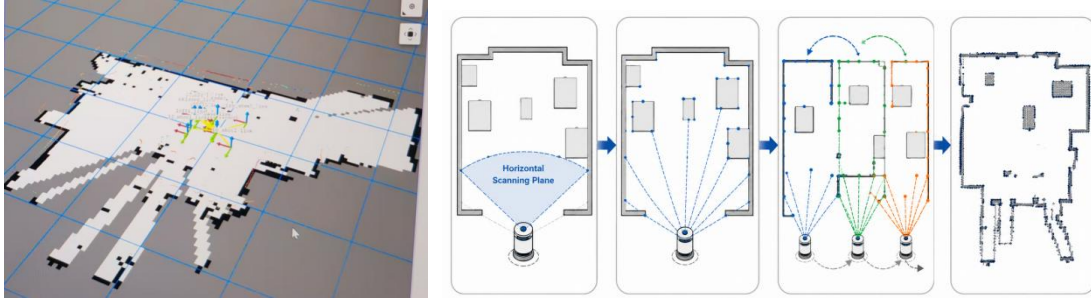


Fig. 6 Scene SLAM Mapping Process

5.2 Module Coordination

The complete system depends on cross-module synchronization. The perception module provides target information, the cognitive module determines the task sequence, the arm controller executes manipulation, and the chassis controller supports movement and alignment. Status feedback from all modules is continuously updated so that the task scheduler can prevent unsafe actions, such as grasping without a valid target or moving while the arm is performing a critical operation.

6. System Testing and Results

The system was tested from perception, cognition, manipulation, and mobility. In the perception test, YOLOv8 recognized target objects in real time, and the RGB-D camera provided usable 3D coordinates for arm control. The microphone array estimated the speaker direction, OCR recognized

package text, and emotion recognition provided interaction-state information. These results verified the feasibility of the multimodal perception design.

In the cognitive interaction test, the robot correctly interpreted natural language commands related to object search, grasping, status inquiry, task execution, and knowledge-based questions. With LangChain and RAG, the system retrieved local knowledge before generating responses or action commands. The structured-command mechanism reduced irrelevant outputs and improved execution safety.



Fig. 7 Dual robotic arm test for drug sorting and water-pouring actions

In the robotic arm test, the system generated feasible inverse-kinematic solutions and smooth planned trajectories for grasping and placing. Dual-arm cooperation was verified in typical service operations. In the mobile platform test, the robot completed basic movement, turning, stopping, and obstacle avoidance using LiDAR data. Overall results show that the robot can complete the closed-loop process from user instruction to perception, reasoning, control, execution, and feedback.

Test Item	Method	Result	Conclusion
Object detection	YOLOv8 visual test	Targets were detected with bounding boxes.	Passed
3D localization	RGB-D conversion	Target coordinates were obtained for grasping.	Passed
Voice interaction	ASR and TTS test	Commands and feedback were completed.	Passed
RAG decision	Language task test	Commands became safe structured actions.	Passed
Arm control	Grasping test	The arm executed planned motions smoothly.	Passed
Navigation	LiDAR obstacle test	The chassis avoided obstacles and adjusted motion.	Passed

7. Key Features and Conclusion

- The system integrates visual, depth, acoustic, textual, and emotional perception into one embodied service robot.
- Large language models are combined with LangChain, FAISS, and RAG to support grounded reasoning and task planning.
- Structured output and rule validation make large-model decisions safer for physical robotic execution.
- Dual-arm manipulation and mobile navigation are integrated, enabling practical service tasks such as grasping, delivery, and assistance.

In conclusion, this project implements a complete intelligent dual-arm robotic system that connects large-model cognition with embedded AI perception and robotic execution. The prototype demonstrates voice interaction, object detection, 3D localization, knowledge retrieval, arm manipulation, obstacle avoidance, and humanized feedback. The work provides a practical reference for applying large language models to embodied robots and can be further extended to elderly care, medical assistance, home service, and indoor delivery applications.